PREMIUM

Full-Length Mock Test

Mock 03

Metallurgical Engineering

GATE MT

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Physical Metallurgy — Diffusion]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.2 [1 Mark] [EASY] [Mechanical Metallurgy — Plasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.3 [1 Mark] [EASY] [Extractive Metallurgy — Hydrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.4 [1 Mark] [EASY] [Physical Metallurgy — Corrosion]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.5 [1 Mark] [EASY] [Mechanical Metallurgy — Plasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.6 [1 Mark] [EASY] [Extractive Metallurgy — Electrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.7 [1 Mark] [EASY] [Physical Metallurgy — Corrosion]

FCC unit cell contains:

A) 1 atom

B) 2 atoms

C) 4 atoms

D) 8 atoms

Q.8 [1 Mark] [EASY] [Mechanical Metallurgy — Fatigue]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.9 [1 Mark] [EASY] [Extractive Metallurgy — Electrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.10 [1 Mark] [EASY] [Physical Metallurgy — Corrosion]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.11 [1 Mark] [EASY] [Mechanical Metallurgy — Plasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.12 [1 Mark] [EASY] [Extractive Metallurgy — Pyrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.13 [2 Marks] [MODERATE] [Physical Metallurgy — Heat Treatment]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.14 [2 Marks] [MODERATE] [Mechanical Metallurgy — Elasticity]

Tempering reduces:

A) Brittleness

B) Hardness

C) Ductility

D) All of the above

Q.15 [2 Marks] [MODERATE] [Extractive Metallurgy — Electrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.16 [2 Marks] [MODERATE] [Physical Metallurgy — Phase Diagrams]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.17 [2 Marks] [MODERATE] [Mechanical Metallurgy — Plasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.18 [2 Marks] [MODERATE] [Extractive Metallurgy — Pyrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.19 [2 Marks] [MODERATE] [Physical Metallurgy — Crystal Structures]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.20 [2 Marks] [MODERATE] [Mechanical Metallurgy — Plasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.21 [2 Marks] [MODERATE] [Extractive Metallurgy — Mineral Processing]

Creep is significant at:

A) Low T

B) High T

C) Room T

D) Cryogenic T

Q.22 [2 Marks] [MODERATE] [Physical Metallurgy — Phase Diagrams]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.23 [2 Marks] [MODERATE] [Mechanical Metallurgy — Creep]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.24 [2 Marks] [MODERATE] [Extractive Metallurgy — Hydrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.25 [2 Marks] [MODERATE] [Physical Metallurgy — Phase Diagrams]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.26 [2 Marks] [MODERATE] [Mechanical Metallurgy — Elasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.27 [2 Marks] [MODERATE] [Extractive Metallurgy — Electrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.28 [2 Marks] [MODERATE] [Physical Metallurgy — Crystal Structures]

FCC unit cell contains:

A) 1 atom

B) 2 atoms

C) 4 atoms

D) 8 atoms

Q.29 [2 Marks] [MODERATE] [Mechanical Metallurgy — Elasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.30 [2 Marks] [MODERATE] [Extractive Metallurgy — Mineral Processing]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.31 [2 Marks] [MODERATE] [Physical Metallurgy — Phase Diagrams]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.32 [2 Marks] [MODERATE] [Mechanical Metallurgy — Elasticity]

Tempering reduces:

A) Brittleness
B) Hardness
C) Ductility
D) All of the above

Q.33 [2 Marks] [HARD] [Extractive Metallurgy — Hydrometallurgy]

Creep is significant at:

A) Low T
B) High T
C) Room T
D) Cryogenic T

Q.34 [2 Marks] [HARD] [Physical Metallurgy — Corrosion]

FCC unit cell contains:

A) 1 atom
B) 2 atoms
C) 4 atoms
D) 8 atoms

Q.35 [2 Marks] [HARD] [Mechanical Metallurgy — Fatigue]

Tempering reduces:

A) Brittleness

B) Hardness

C) Ductility

D) All of the above

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Extractive Metallurgy — Pyrometallurgy]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Physical Metallurgy — Heat Treatment]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Mechanical Metallurgy — Fracture]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Extractive Metallurgy — Mineral Processing]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Physical Metallurgy — Corrosion]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Extractive Metallurgy — Pyrometallurgy]

FCC has how many atoms per unit cell?

Answer: ____________________

Q.52 [2 Marks] [HARD] [Physical Metallurgy — Corrosion]

Carbon steel 0.8% C is:

Answer: ____________________

Q.53 [2 Marks] [HARD] [Mechanical Metallurgy — Fatigue]

FCC has how many atoms per unit cell?

Answer: ____________________

Q.54 [2 Marks] [HARD] [Extractive Metallurgy — Mineral Processing]

Carbon steel 0.8% C is:

Answer: ____________________

Q.55 [2 Marks] [HARD] [Physical Metallurgy — Phase Diagrams]

FCC has how many atoms per unit cell?

Answer: ____________________

Q.56 [2 Marks] [HARD] [Mechanical Metallurgy — Fracture]

Carbon steel 0.8% C is:

Answer: ____________________

Q.57 [2 Marks] [HARD] [Extractive Metallurgy — Pyrometallurgy]

FCC has how many atoms per unit cell?

Answer: ____________________

Q.58 [2 Marks] [HARD] [Physical Metallurgy — Corrosion]

Carbon steel 0.8% C is:

Answer: ______________________

Q.59 [2 Marks] [HARD] [Mechanical Metallurgy — Fatigue]

FCC has how many atoms per unit cell?

Answer: ____________________________

Q.60 [2 Marks] [HARD] [Extractive Metallurgy — Electrometallurgy]

Carbon steel 0.8% C is:

Answer: ____________________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	C	**Q.2:**	D	**Q.3:**	B
Q.4:	C	**Q.5:**	D	**Q.6:**	B
Q.7:	C	**Q.8:**	D	**Q.9:**	B
Q.10:	C	**Q.11:**	D	**Q.12:**	B
Q.13:	C	**Q.14:**	D	**Q.15:**	B
Q.16:	C	**Q.17:**	D	**Q.18:**	B
Q.19:	C	**Q.20:**	D	**Q.21:**	B
Q.22:	C	**Q.23:**	D	**Q.24:**	B
Q.25:	C	**Q.26:**	D	**Q.27:**	B
Q.28:	C	**Q.29:**	D	**Q.30:**	B
Q.31:	C	**Q.32:**	D	**Q.33:**	B
Q.34:	C	**Q.35:**	D	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	4
Q.52:	4	**Q.53:**	4	**Q.54:**	4
Q.55:	4	**Q.56:**	4	**Q.57:**	4
Q.58:	4	**Q.59:**	4	**Q.60:**	4

Detailed Solutions

Question 1 *[Physical Metallurgy — Diffusion]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 2 *[Mechanical Metallurgy — Plasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 3 *[Extractive Metallurgy — Hydrometallurgy]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 4 *[Physical Metallurgy — Corrosion]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 5 *[Mechanical Metallurgy — Plasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 6 *[Extractive Metallurgy — Electrometallurgy]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 7 *[Physical Metallurgy — Corrosion]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 8 *[Mechanical Metallurgy — Fatigue]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 9 *[Extractive Metallurgy — Electrometallurgy]*

Creep is significant at:

Answer:

(B)

Creep is time-dependent deformation significant at high temperatures.

Question 10 *[Physical Metallurgy — Corrosion]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 11 *[Mechanical Metallurgy — Plasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 12 *[Extractive Metallurgy — Pyrometallurgy]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 13 *[Physical Metallurgy — Heat Treatment]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 14 *[Mechanical Metallurgy — Elasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 15 *[Extractive Metallurgy — Electrometallurgy]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 16 *[Physical Metallurgy — Phase Diagrams]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 17 *[Mechanical Metallurgy — Plasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 18 *[Extractive Metallurgy — Pyrometallurgy]*

Creep is significant at:

Answer:

(B)

Creep is time-dependent deformation significant at high temperatures.

Question 19 *[Physical Metallurgy — Crystal Structures]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 20 *[Mechanical Metallurgy — Plasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 21 *[Extractive Metallurgy — Mineral Processing]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 22 *[Physical Metallurgy — Phase Diagrams]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 23 *[Mechanical Metallurgy — Creep]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 24 *[Extractive Metallurgy — Hydrometallurgy]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 25 *[Physical Metallurgy — Phase Diagrams]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 26 *[Mechanical Metallurgy — Elasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 27 *[Extractive Metallurgy — Electrometallurgy]*

Creep is significant at:

Answer:

(B)

Creep is time-dependent deformation significant at high temperatures.

Question 28 *[Physical Metallurgy — Crystal Structures]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 29 *[Mechanical Metallurgy — Elasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 30 *[Extractive Metallurgy — Mineral Processing]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 31 *[Physical Metallurgy — Phase Diagrams]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 32 *[Mechanical Metallurgy — Elasticity]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 33 *[Extractive Metallurgy — Hydrometallurgy]*

Creep is significant at:

Answer: (B)

Creep is time-dependent deformation significant at high temperatures.

Question 34 *[Physical Metallurgy — Corrosion]*

FCC unit cell contains:

Answer: (C)

FCC: 8 corners × 1/8 + 6 faces × 1/2 = 4 atoms.

Question 35 *[Mechanical Metallurgy — Fatigue]*

Tempering reduces:

Answer: (D)

Tempering reduces brittleness, hardness, and improves ductility.

Question 36 *[Extractive Metallurgy — Pyrometallurgy]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37 *[Physical Metallurgy — Heat Treatment]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 *[Mechanical Metallurgy — Fracture]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 *[Extractive Metallurgy — Mineral Processing]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 *[Physical Metallurgy — Corrosion]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 *[Mechanical Metallurgy — Fatigue]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 *[Extractive Metallurgy — Electrometallurgy]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 *[Physical Metallurgy — Heat Treatment]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 *[Mechanical Metallurgy — Fracture]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 *[Extractive Metallurgy — Pyrometallurgy]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Physical Metallurgy — Phase Diagrams]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Mechanical Metallurgy — Plasticity]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Extractive Metallurgy — Mineral Processing]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Physical Metallurgy — Phase Diagrams]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Mechanical Metallurgy — Plasticity]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Extractive Metallurgy — Pyrometallurgy]*

FCC has how many atoms per unit cell?

Answer: (4)

FCC: 8 corners×1/8 + 6 faces×1/2 = 4 atoms.

Question 52 *[Physical Metallurgy — Corrosion]*

Carbon steel 0.8% C is:

Answer: (4)

0.8% C is exactly eutectoid composition.

Question 53 *[Mechanical Metallurgy — Fatigue]*

FCC has how many atoms per unit cell?

Answer: (4)

FCC: 8 corners×1/8 + 6 faces×1/2 = 4 atoms.

Question 54 *[Extractive Metallurgy — Mineral Processing]*

Carbon steel 0.8% C is:

Answer:

(4)

0.8% C is exactly eutectoid composition.

Question 55 *[Physical Metallurgy — Phase Diagrams]*

FCC has how many atoms per unit cell?

Answer: (4)

FCC: 8 corners×1/8 + 6 faces×1/2 = 4 atoms.

Question 56 *[Mechanical Metallurgy — Fracture]*

Carbon steel 0.8% C is:

Answer: (4)

0.8% C is exactly eutectoid composition.

Question 57 *[Extractive Metallurgy — Pyrometallurgy]*

FCC has how many atoms per unit cell?

Answer: (4)

FCC: 8 corners×1/8 + 6 faces×1/2 = 4 atoms.

Question 58 *[Physical Metallurgy — Corrosion]*

Carbon steel 0.8% C is:

Answer: (4)

0.8% C is exactly eutectoid composition.

Question 59 *[Mechanical Metallurgy — Fatigue]*

FCC has how many atoms per unit cell?

Answer: (4)

FCC: 8 corners×1/8 + 6 faces×1/2 = 4 atoms.

Question 60 *[Extractive Metallurgy — Electrometallurgy]*

Carbon steel 0.8% C is:

Answer: (4)

0.8% C is exactly eutectoid composition.